EITQ TD: Quantum(-related) complexity

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1 QMA-Completeness

We consider here two decision problems:

1. **Non-Identity Check**
   Setup: A quantum circuit C that implements an (unknown) unitary U.
   Question: Does $U \neq e^{\mathbf{i}\phi}\mathrm{Id}$ for all ϕ?

2. **Non-Equivalence Check**
   Setup: Two quantum circuits C_1 and C_2 that implement respectively U_1 and U_2.
   Question: Does $U_1 \neq e^{\mathbf{i}\phi}U_2$ for all ϕ?

Question 1. Is one of the two problems a sub-problem of the other?

Non-Identity Check is a subproblem of Non-Equivalence Check. It suffices to set C_2 to the identity circuit in the latter to recover the former.

Question 2. Provide a polynomial reduction from the non-equivalence check problem to the non-identity check problem.

Build the circuit $C_1 \circ C_2^\dagger$. This can be done in polynomial time in the size of the two circuits. Then, C_1 and C_2 are equivalent iff $C_1 \circ C_2^\dagger$ is the identity (up to a phase). We hence have a reductionfrom the non-equivalence check problem to the non-identity check problem.

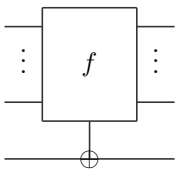
Question 3. Assuming the non-identity check problem is QMA-complete, show that the Non-Equivalence check problem is also QMA-complete.

From the first two questions, we have a polynomial reduction from each of the two problems to the other, hence they are equivalent. Since the first is QMA-complete, so is the second.

2 Hermiticity Checking is coNP-hard

We are interested here in the following problem:
Setup: A circuit C built using gate set $\langle H, P(\alpha), \mathrm{CNot}\rangle_{\alpha\in\mathbb{R}}$.
Question: Does C implement a Hermitian matrix?

Let f be a SAT formula. We define circuit C_f, represented as:



and which maps classical data as follows:

$$U_f|x_1, ..., x_n, y\rangle \mapsto |x_1, ..., x_n, y \oplus f(x_1, ..., x_n)\rangle$$

Question 1. Show that matrix U_f is block-diagonal (when taking the usual convention that A above B represents $A \otimes B$). What are the blocks on the diagonal, depending on the values of f?

With X the Pauli matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, the matrix U_f can be written:

$$\begin{pmatrix} X^{f(0,\dots,0,0)} & & & \\ & X^{f(0,\dots,0,1)} & & \\ & & \ddots & \\ & & & X^{f(1,\dots,1)} \end{pmatrix}$$

Question 2. What is the matrix implemented by the circuit C'_f built from C_f by adding a Pauli Z gate on the last qubit at the end?

The matrix becomes:

$$\begin{pmatrix} ZX^{f(0,\dots,0,0)} & & & \\ & ZX^{f(0,\dots,0,1)} & & \\ & & \ddots & \\ & & & ZX^{f(1,\dots,1)} \end{pmatrix}$$

Question 3. Is Z Hermitian? Is ZX Hermitian? Then show that C'_f implements a Hermitian matrix iff f has no solution.

Z is Hermitian (i.e. $Z^\dagger = Z$, so are all Pauli matrices). ZX is not Hermitian, the two Pauli matrices anticommute: $(ZX)^\dagger = X^\dagger Z^\dagger = XZ = -ZX$. For a block diagonal matrix to be Hermitian, all its blocks have to be Hermitian. Hence, the above matrix is Hermitian iff f is constant to 0, i.e. iff f has no solution.

Question 4. Assuming that the unsatisfiability problem UNSAT (deciding if f has no solution) is **coNP-complete**; and that C_f can be built from f using a number of gates drawn from $\langle H, P(\alpha), \text{CNot}\rangle_{\alpha\in\mathbb{R}}$ that is polynomial in the size of f, show that Hermiticity-checking is **coNP-hard**.

We have a polynomial reduction from the UNSAT problem to the Hermiticity Checking problem. Indeed the above circuit is built in polynomial time (and size) in the size of the formula f, and it is Hermitian iff f has is unsatisfiable. UNSAT being **coNP-complete**, this proves that Hermiticity Checking is **coNP-hard**.

Question 5. What can we say about the complexity of checking whether a circuit C implements an involutive matrix (i.e. $U \circ U = \text{Id}$)?

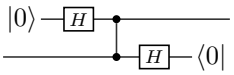
Since U is implemented by a circuit, it is unitary. It turns out that for unitary maps, involutivity is equivalent to Hermiticity:

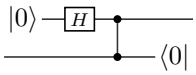
$$U \circ U = Id \iff U^\dagger \circ U \circ U = U^\dagger \iff U = U^\dagger$$

Hence the complexity of that problem is the same, it is **coNP-hard**.

3 Unitarity Checking is coNP-hard

Question 1. Compute the operators implemented by the following circuits with postselections:





Are they unitary?

The first circuit implements the following map:

$$(\langle 0|\otimes I)(H\otimes I)CZ(I\otimes H)(I\otimes|0\rangle) = (\langle +|\otimes I)CZ(I\otimes|+\rangle) = \frac{1}{2}\begin{pmatrix}1&0&1&0\\0&1&0&1\end{pmatrix}\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&-1\end{pmatrix}\begin{pmatrix}1&0\\1&0\\0&1\\0&1\end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix}1&1\\1&-1\end{pmatrix}$$

Up to renormalisation (which occurs after the postselection), the circuit implements the Hadamard gate, which is unitary.

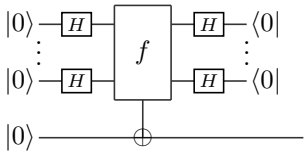
The second circuit implements:

$$(\langle 0|\otimes I)(H\otimes I)CZ(I\otimes|0\rangle) = |+\rangle\langle 0|$$

which is not unitary.

Let f be a SAT formula on n variables, and s be the number of variable assignments that satisfy f. We first want to encode the s into a quantum state $|\psi\rangle$ (using postselections).

Question 2. Show that the following circuit implements $|\psi\rangle := \frac{1}{N}\big((2^n-s)|0\rangle + s|1\rangle\big)$:



After the first round of H gates, we get state:

$$\frac{1}{\sqrt{2}^n}\sum_{\vec{x}\in\{0,1\}^n}|\vec{x},0\rangle$$

After the f oracle, we get:

$$\frac{1}{\sqrt{2}^n}\sum_{\vec{x}\in\{0,1\}^n}|\vec{x},f(\vec{x})\rangle$$

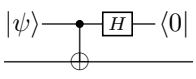
The next round of H gates and postselections modifies the state as follows:

$$\left(\frac{1}{\sqrt{2}^n}\sum_{\vec{y}\in\{0,1\}^n,z\in\{0,1\}}|z\rangle\langle\vec{y},z|\right)\left(\frac{1}{\sqrt{2}^n}\sum_{\vec{x}\in\{0,1\}^n}|\vec{x},f(\vec{x})\rangle\right) = \frac{1}{2^n}\sum_{\vec{x}\in\{0,1\}^n}|f(\vec{x})\rangle$$

With s the number of solutions to f, i.e. the number of values for $\vec{x}\in\{0,1\}^n$ such that $f(\vec{x})=1$, the state is then, up to renormalisation:

$$|\psi\rangle := \frac{1}{N}\big((2^n-s)|0\rangle + s|1\rangle\big)$$

Question 3. What operator is implemented by the following circuit:



Check that it is unitary iff $s=0$ or $s=2^n$.

This operator is:

$$(\langle +| \otimes I)CX(|\psi\rangle \otimes I) = \frac{1}{\sqrt{2}N}\begin{pmatrix}1&0&1&0\\0&1&0&1\end{pmatrix}\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}\begin{pmatrix}2^n-s&0\\0&2^n-1\\s&0\\0&s\end{pmatrix} = \frac{1}{\sqrt{2}N}\begin{pmatrix}2^n-s&s\\s&2^n-s\end{pmatrix}$$

Postselection renormalises the operator, so this operator is unitary if its composition with its dagger is the identity up to a global factor. This composition amounts to:

$$\begin{pmatrix}2^n-s&s\\s&2^n-s\end{pmatrix}\begin{pmatrix}2^n-s&s\\s&2^n-s\end{pmatrix} = \begin{pmatrix}(2^n-s)^2+s^2&2s(2^n-s)\\2s(2^n-s)&(2^n-s)^2+s^2\end{pmatrix}$$

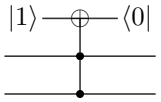
We then easily see that the result is colinear to the identity iff $s = 0$ or $s = 2^n$.

Question 4. Using the fact that UNSAT is a **coNP-complete** problem, conclude on the hardness of checking unitarity of circuits with postselections.
Hint: don't forget to deal with the $s = 2^n$ case.

Let f be a SAT formula, we are interested in the unsatisfiability of that formula. We can first check $f(0, ..., 0)$. If the result is 1, we can stop and say that f is not unsatisfiable. If the result is 0, we have ruled out the $s = 2^n$ case, and we can proceed with the composition of the two circuits from the two previous questions. The resulting circuit (with postselections) is of polynomial size in the size of f, and it implements a unitary iff f has no solution, i.e. if it is unsatisfiable. We hence have a polynomial reduction from UNSAT to Unitarity Check. UNSAT being **coNP-complete**, Unitarity Check is hence **coNP-hard**.

4 The Power of a Single Postselection

Question 1. Explain the behaviour of the following circuits (compute what they implement if you need), and provide an equivalent simplified circuit for each:





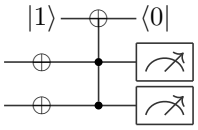
With x and y the (classical) inputs to the first circuit, it performs the following:

$$|x, y\rangle \mapsto |x, y, 1\rangle \mapsto |x, y, 1 \oplus xy\rangle \mapsto |x, y\rangle\langle 0|1 \oplus xy\rangle = \delta_{x,1}\delta_{y,1}|1, 1\rangle$$

So, on classical inputs, it forces them to be in the 1 state. By linearity, on linear combinations of classical inputs, it only keeps the terms where the two inputs are in state 1.

The second circuit creates a qubit is state $|1\rangle$, and then measures it. This piece of circuit is hence useless.

Question 2. Use the above to show that the following circuit is equivalent to 2 postselections in $\langle 0|$:



After the postselection, but before the measurements, the circuit does the following to classical states x and y:

$$|x, y\rangle \mapsto \delta_{x,0}\delta_{y,0}|1, 1\rangle$$

The measurements then get rid of the two qubits (as does the second circuit in the previous question). Hence, the full circuit does $|x, y\rangle \mapsto \delta_{x,0}\delta_{y,0}$ on classical data, i.e. its interpretation is $\langle 0, 0|$, the same as two postselections in $\langle 0|$.

Question 3. Show that having a single postselection is as strong as having arbitrarily many postselections.

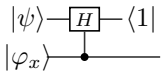
Starting with a circuit with many postselections, we can do the following: as long as we have at least 2 of them, replace them with the circuit from the previous question. In the end, we are left with a single postselection, and the resulting circuit is of linear size in the size of the original one. Alternatively, one can generalise the above circuit to accept arbitrarily many qubits (using a multi-control X gate), and replace all postselections at once with it.

5 PP $\subseteq$ PostBQP

Let f be a SAT formula on n variables, and s be the number of variable assignments that satisfy f. The majority problem asks whether $s < 2^{n-1}$ or $s \geq 2^{n-1}$.

We assume we know how to create state $|\psi\rangle$ from f as in Question 2 from Exercise 3.

Question 1. Let $|\varphi_x\rangle$ be the arbitrary state $\frac{1}{N}(|0\rangle + x|1\rangle)$ where N is a normalisation factor. Compute the state $|\psi_x\rangle$ created by the following circuit:



We start with $|\psi\rangle \otimes |\varphi_x\rangle = \frac{1}{N'}((2^n - s)|00\rangle + s|10\rangle + x(2^n - s)|01\rangle + xs|11\rangle$. After application of the controlled-H gate, we have:

$$\frac{1}{N'}\left((2^n - s)|00\rangle + s|10\rangle + \frac{x(2^n - s) + xs}{\sqrt{2}}|01\rangle + \frac{x(2^n - s) - xs}{\sqrt{2}}|11\rangle\right)$$

After postselection of the first qubit in $|1\rangle$, we have:

$$|\psi_x\rangle = \frac{1}{N''}\left(s|0\rangle + \frac{x(2^n - 2s)}{\sqrt{2}}|1\rangle\right)$$

We then reason w.r.t. the value of s.

Question 2. If $s < 2^{n-1}$, explain which i maximises $\langle +|\psi_{2^i}\rangle$. Notice that for such i, if $|\psi_{2^i}\rangle = \alpha|0\rangle + \beta|1\rangle$, then α is within $\beta/2$ and 2β. Infer a lower bound on $\langle +|\psi_{2^i}\rangle$.

If $s < 2^{n-1}$, and $x = 2^i$, then both amplitudes are positive. To maximise $\langle +|\psi_{2^i}\rangle$ we want both amplitudes to be as close as possible, hence ideally $i = \log_2\left(\frac{s\sqrt{2}}{2^n - 2s}\right)$. i has to be an integer, so we simply round this value to get the best value.

With the above observation, $\langle +|\psi_{2^i}\rangle$ has to be at least as big as both $\langle +|\varphi_2\rangle$ and $\langle +|\varphi_{1/2}\rangle$, both of which are $\frac{3}{\sqrt{10}} \geq 0.948$. Hence $\langle +|\psi_{2^i}\rangle \geq 0.948$.

Question 3. If $s \geq 2^{n-1}$, show that for all $i \in [\![-n, n]\!]$, $|\langle +|\psi_{2^i}\rangle| \leq \frac{1}{\sqrt{2}} \simeq 0.707$.

If $s \geq 2^{n-1}$, for all values of i, the second amplitude of $|\psi_{2^i}\rangle$ is non-positive while the first is positive. There exists θ such that $|\psi_{2^i}\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle$. Then:

$$|\langle +|\psi_{2^i}\rangle| = \frac{1}{\sqrt{2}}|\cos\theta - \sin\theta| \leq \frac{1}{\sqrt{2}} \simeq 0.707$$

Question 4. Using the above, come up with a polynomial time algorithm that decides the majority problem. Conclude on the interaction between **PP** and **PostBQP**.

If $s < 2^{n-1}$, there exists $i \in [-n, n]$ such that $|\langle +|\psi_{2^i}\rangle|^2 \geq 0.9$, however if $s \geq 2^{n-1}$ then for all i, $|\langle +|\psi_{2^i}\rangle|^2 \leq 0.5$. For each i between $-n$ and n, we build the state $|\psi_{2^i}\rangle$ and measure it in the X basis. We do so n times, so that we can discriminate between the two possibilities with exponentially good certainty. We hence run the circuit $n(2n+1)$ times, i.e. we have a polynomial-sized algorithm with postselections that solve the majority problem.

Since this problem is **PP-complete** and we have a polynomial reduction from this problem to a **Post-BQP** problem, we can conclude that **PP** $\subseteq$ **PostBQP**.